

When Do AI-powered Service Chatbots Feel Natural? The Role of Conversational Embodiment and Relational Orientation

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Abstract

Building on Communication Accommodation Theory, this study examines the effectiveness of conversational embodiment in AI-powered service chatbots-encompassing turn-taking, turn-initiating, and grounding-in shaping service performance, with relational orientation as a key boundary condition. Two experimental studies validate a serial mediation model in which conversational embodiment enhances perceived naturalness, which subsequently improves communication quality and service satisfaction. The moderating role of relational orientation is further confirmed that for communally oriented users, higher levels of conversational embodiment-particularly grounding- significantly increase perceived naturalness and satisfaction. In contrast, for exchange-oriented users who prioritize task efficiency, these effects are relatively weak or non-significant. These findings underscore the limitations of a one-size-fits-all approach and highlight the need to tailor the design of AI-powered conversational agents to the relational norm of the target consumers.

CCS Concepts

• **Human-centered computing** → **Empirical studies in interaction design**.

Keywords

AI-powered chatbot, Conversational embodiment, Naturalness, Exchange orientation, Communal orientation

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1 Introduction

As AI service technologies increasingly acquire interactive and human-like communication capabilities, elements that reflect human communication patterns are emerging as core requirements for interface design. While previous research has focused on physical design cues like gaze or tone of voice, this study suggests that the dynamics of the conversational interaction itself, specifically

conversational embodiment, play a central role in shaping service experiences.

Utilizing Communication Accommodation Theory (CAT) as a foundational framework, this study investigates the dynamics of conversational interactions and how users evaluate them based on an AI's ability to mirror the flow, pacing, and structure of human dialogue. Of primary interest in this evaluation is perceived naturalness, which represents a user's assessment of whether the chatbot's responses align with established conversational norms, like appropriate grounding and turn-taking. These perceptions, in turn, shape a comprehensive assessment of communication quality, encompassing the overall clarity, fluidity, and efficacy of the interaction.

The purpose of this research is to investigate how the design of AI-powered chatbots, specifically their level of conversational embodiment, influences user service satisfaction. Moving beyond physical anthropomorphism, this study aims to determine if the positive impact of conversational cues like turn-taking and grounding on service satisfaction is driven by perceived naturalness and communication quality. Furthermore, the research intends to identify critical boundary conditions by examining how individual differences in relational orientation (communal vs. exchange) moderate a user's sensitivity to these cues. By exploring these dynamics, the study provides a nuanced understanding of when human-like dialogue patterns are most effective, offering a more tailored approach to AI-powered service design.

2 Theoretical Background

2.1 Conversational Embodiment

Sociolinguistics and human interaction scholars describe conversation not as a simple exchange of information, but as a social coordination process aimed at forming shared understanding [5]. Individuals negotiate social cues through conversational embodiment, fostering mutual coordination and relational engagement [9]. Prior work has emphasized the role of human-like linguistic patterns in shaping how AI-mediated conversations are perceived in service settings [4, 18]. Rather than relying on physical forms of embodiment, such as robotic or avatar appearances, verbal embodiment is realized through linguistic coordination and the sequential organization of dialogue [4, 18].

Building on this perspective, Bergner et al. [4] conceptualized conversational embodiment a continuum of engagement that encompasses turn-taking, turn-initiating, and grounding. Turn-taking and turn-initiating shape the reciprocity and progression of interaction and are often jointly enacted in conversational design rather than operating as independent mechanisms[15]. Grounding,

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in turn, facilitates conversational continuity by signaling comprehension and acceptance through verbal cues[4]. As these linguistic patterns become more pronounced, conversational agents are increasingly perceived as socially responsive, which, in turn, shapes users' evaluative responses to the interaction[4].

2.2 Communication Accommodation Theory: Perceived Naturalness and Communication Quality

Communication Accommodation Theory (CAT) provides a useful theoretical lens for explaining how conversational interactions unfold and how they can be evaluated [11]. The CAT perspective clarifies how such perceptions emerge through the dynamics of conversational interaction [4, 11]. The theory explains a dynamic process in which conversational participants adjust their linguistic styles and modes of expression to foster mutual understanding and relational coherence [10]. Such adjustments signal a willingness to engage in communication with the other party and serve as a key factor in shaping the overall evaluation of the conversational experience [11]. When this perspective is applied to AI-powered conversational services, it suggests that users do not assess the interaction solely based on the usefulness of the AI task [2]. Rather, they also attend to how appropriately the AI adapts to the flow, pacing, and structure of human conversation [2]. Within this context, perceived naturalness captures users' judgments of whether the AI's responses conform to expected conversational norms, such as appropriate turn-taking and grounding, within the given situation. These judgments then inform a broader, more integrative evaluation of the interaction, commonly conceptualized as communication quality, which reflects the overall clarity, smoothness, and effectiveness of the conversational exchange [14]. Taken together, perceived naturalness and communication quality function as key evaluative mechanisms through which experiences with AI-based service agents are translated into overall service satisfaction.

2.3 Relational Orientation as a Moderator: Exchange vs. Communal Orientation

Importantly, users' perceptions toward communication interactions with AI-based conversational chatbots may vary depending on users' relational orientation. According to communal and exchange relationship theory, individuals differ in how they approach and interpret interactions[7]. Specifically, in AI-based service environments, communally oriented individuals may be more sensitive to interactional cues inherent in conversations, such as how responses connect and build upon the ongoing conversation [8]. In contrast, exchange-oriented users may focus less on such interactional sophistication and more on whether the conversation effectively supports task performance [8].

Given these differences, the presence of conversational embodiment (i.e., social cues) is expected to generate in how natural the interaction feels and how effective the communication is perceived across users, as relational norms shape the interpretation of social and interactional cues [1]. In this sense, recent research on conversational AI demonstrates that social cues embedded in dialogue play a critical role in shaping users' evaluations of interaction quality[4].

Based on this reasoning, this study predicts that the effect of conversational embodiment in AI-based conversational agents on service satisfaction will be sequentially mediated by perceived naturalness and communication quality (H1). Moreover, this indirect effect is expected to vary as a function of users' relational orientation. Specifically, communally oriented users are likely to exhibit a stronger positive relationship between conversational embodiment in AI-based chatbot interfaces and perceived naturalness than exchange-oriented users (H2). Thus, we suggest following hypotheses.

- **H1.** The positive effect of the AI-powered conversational agents—specifically, higher levels of conversational embodiment (grounding relative to turn-taking and baseline)—on service satisfaction is serially mediated by perceived naturalness and communication quality.
- **H2.** Relational orientation moderates the effect of the AI-powered conversational agents—specifically, higher levels of conversational embodiment (grounding relative to turn-taking and baseline)—on perceived naturalness, such that the positive effect is stronger for communally oriented consumers (vs. exchange-oriented consumers).

2.4 Relational Orientation as a Moderator of the Serial Mediation Process

Taken together, relational orientation not only plays a significant role in shaping users' perceptions of the human-likeness of the conversation (i.e., naturalness) but also influences the overall process by which this perception translates into subsequent evaluations toward communication quality [1, 4, 6]. As community-oriented users are highly sensitive to interactional and relational cues in conversations, increasing the conversational embodiment (i.e., social cues through turn-taking/grounding) is likely to lead to a more natural perception of interactions, which in turn, is likely to lead to a more positive evaluation of communication quality [16]. These sequential dynamics may enhance overall satisfaction with AI-based services. Conversely, exchange-oriented users tend to focus more on task efficiency than relational sophistication of conversations [6, 13]. Therefore, the indirect effect of the conversational cues on service satisfaction via perceived naturalness and communication quality may be relatively weak [16]. Therefore, relational orientation is expected to moderate the overall indirect effect, which sequentially mediates the impact of conversational implementation on service satisfaction through perceived naturalness and communication quality.

- **H3.** Relational orientation moderates the serial indirect effect of the AI-powered conversational agents—operationalized as increasing levels of conversational embodiment (grounding relative to turn-taking and baseline)—on service satisfaction via perceived naturalness and communication quality, such that the serial indirect effect is stronger for communally oriented consumers (vs. exchange-oriented consumers).

3 Method

To test the proposed hypotheses, this research employed experimental studies. Study 1 tested the serial mediation model using a between-subjects factorial design with three levels of AI-powered

conversational agency (baseline vs. turn-taking vs. grounding). The study examines whether the effect of conversational embodiment on service satisfaction is serially mediated by perceived naturalness and communication quality (H1). Study 2 extended this model by incorporating relational orientation as a boundary condition, employing a 3 (AI-powered conversational agency: baseline vs. turn-taking vs. grounding) $\times$ 2 (relational orientation: exchange vs. communal) between-subjects factorial design. This design allowed us to test both the interaction between AI-powered conversational agency and relational orientation on perceived naturalness (H2) and the resulting moderated serial mediation effect on service satisfaction (H3).

3.1 Pretest: Development of AI Conversational Agency Conditions

A pretest using a scenario-based online experiment was conducted to validate the manipulation of three AI chatbot interaction conditions—baseline, turn-taking, and grounding—which were developed to differ in their level of conversational embodiment. The baseline chatbot was designed to be minimally conversational, presenting all questions at once without interactive feedback. The turn-taking condition enabled reciprocal exchange by presenting questions sequentially, allowing alternating user-system turns. The grounding condition built upon the turn-taking structure by incorporating explicit acknowledgment and confirmation cues (e.g., “Understood,” “Thanks for sharing that”) to signal shared understanding. A total of 160 MTurk participants (M age = 34.2; 44% female; 82.3% White) completed the pretest. Participants randomly assigned to interact with one of three chatbot interfaces. During the conversational interaction, participants engaged in a real-time, turn-by-turn conversation with a chatbot in a simulated shopping context. After completing the interaction, perceived turn-taking and perceived grounding were assessed [4]. A MANOVA indicated a significant multivariate effect of interaction type: $F(4, 312) = 5.09, p < .001$. Follow-up ANOVAs indicated that perceived turn-taking was significantly lower in the baseline condition than in the turn-taking and grounding conditions, and perceived grounding was highest in the grounding condition ($ps < .01$). Scenario realism did not differ across conditions, confirming the effectiveness of the interaction-type manipulation.

3.2 Procedure and Data Analyses

A total of 165 MTurk participants were recruited for Study 1 (M age = 32.7 years, 36% female; 83.6% White). Participants completed the study and were randomly assigned to interact with one of three AI-powered chatbot interfaces: baseline, turn-taking, or grounding. After completing the interaction, participants responded to measures assessing perceived naturalness [3], communication quality [17] and service satisfaction [18].

A total of 169 MTurk participants were recruited for Study 2 (M age = 33 years; 52.4% male; 83.1% White). Study 2 followed the same procedure and employed the same simulated shopping scenario used in Study 1. First, relational orientation was assessed prior to the chatbot interaction using separate continuous scales for exchange orientation (4 items) and communal orientation (3 items) [7], and

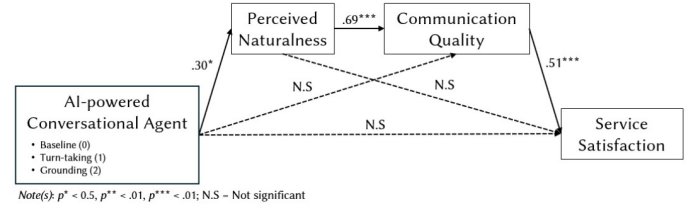


Figure 1: Research model and path results (Study 1)

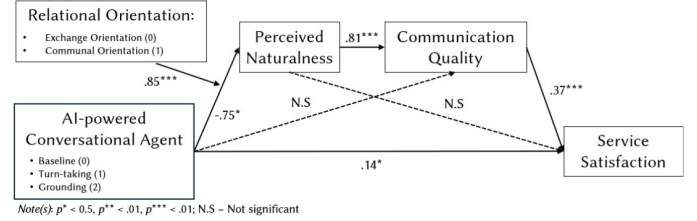


Figure 2: Research model and path results (Study 2)

participants were classified into exchange-oriented and communal-oriented groups using a median-split approach based on their dominant relational orientation. Participants were randomly assigned one of three chatbot interfaces to engage in a real-time conversation with a chatbot. Following the interaction, they completed the measures of perceived naturalness, communication quality, and service satisfaction (used in study 1).

Study 1 employed a three-condition (AI-powered conversational agent: baseline vs. turn-taking vs. grounding) between-subjects factorial design and tested a serial mediation model (PROCESS Model 6) [12], where the conversational agent as the independent variable, perceived naturalness and communication quality as sequential mediators, and service satisfaction as the dependent variable. Study 2 employed a 2 (relational orientation: exchange vs. communal) $\times$ 3 (conversational agent type) between-subjects factorial design and tested moderated serial mediation (PROCESS Model 83) [14], where relational orientation was adopted as a moderator.

4 Results

4.1 Study 1

Results (Figure 1) showed that the AI-powered conversational agent significantly increased perceived naturalness ($B = .30, t = 2.35, p < .05, 95\%CI[0.05, 0.55]$). Perceived naturalness, in turn, significantly explained communication quality ($B = .69, t = 18.33, p < .001, 95\%CI[0.61, 0.76]$). Communication quality also has a significant positive effect on service satisfaction ($B = .51, t = 8.75, p < .001, 95\%CI[0.40, 0.63]$). The direct effect of conversational agent type on service satisfaction was not significant ($B = .06, t = 1.27, p = .21, 95\%CI[-0.03, 0.15]$). Regarding H1, the sequential moderate effect of AI-powered conversational agent on service satisfaction through perceived naturalness and communication quality was significant (indirect effect = .10, 95% Boot CI [.01, .23]). Thus, H1 was supported.

4.2 Study 2

Study 2 examined whether relational orientation (0 = exchange orientation, 1 = communal orientation) moderates the effect of conversational agent type on perceived naturalness and the resulting serial indirect effect on service satisfaction (PROCESS model 83; Figure 2). Results showed a significant interaction between conversational agent type and relational orientation on perceived naturalness ($B = .85$, $t = 3.84$, $p < .001$, 95% CI [0.41, 1.29]). Regarding the interaction effect, conditional effects indicated that for communal-oriented participants, conversational agent type had a strong positive effect on perceived naturalness ($B = .95$, $t = 5.89$, $p < .001$, 95% CI [0.63, 1.27]). In contrast, this effect was not significant for exchange-oriented participants ($B = .10$, $t = 0.65$, $p = .51$, 95% CI [-0.20, 0.40]). Figure 3 indicates that perceived naturalness increases with higher levels of conversational embodiment (turn-taking and grounding) among communally oriented participants, whereas minimal variation is observed for exchange-oriented participants. Thus, H2 was accepted. Regarding H3, we found significant moderated serial mediation effect predicting service satisfaction via perceived naturalness and communication quality (index = .25, 95% Boot CI [0.06, 0.50]). Consistent with expectations, the conditional indirect effect of conversational agent type on service satisfaction through perceived naturalness and communication quality was significant for communally oriented participants (indirect effect = .28, 95% Boot CI [0.08, 0.53]), but not for exchange-oriented participants (indirect effect = .03, 95% Boot CI [-0.01, 0.09]). Thus, H3 was accepted.

5 Discussion and Conclusion

The results of this study indicate that higher levels of conversational embodiment in AI-based conversational agents—particularly when social cues such as turn-taking and grounding are more implemented—contribute to service satisfaction through a sequential increase in perceived naturalness and communication quality. Findings of this study also suggest that these conversational design features do not exert uniform effects across users. Instead, their influence unfolds through different cognitive and evaluative pathways depending on their relational orientation.

This study expands the CAT framework by positioning relational orientation as a key moderating variable, that shapes how conversational embodiment is translated into evaluative outcomes toward the AI service. Furthermore, by integrating community and exchange relationship theories with conversational AI research, this study explains how and why interactive conversational dynamics enhance perceived naturalness and communication quality for communally oriented consumers but have relatively limited for exchange-oriented consumers.

From a practical perspective, the results underscore the limitations of a one-size-fits-all approach to designing AI-based conversational agents. In service settings that primarily engage communally oriented consumers, enhancing conversational cues (i.e., contextual grounding), can meaningfully improve service quality. Conversely, for exchange-oriented consumers, design strategies prioritizing task efficiency without adding unnecessary interactional complexity are likely to be more appropriate.

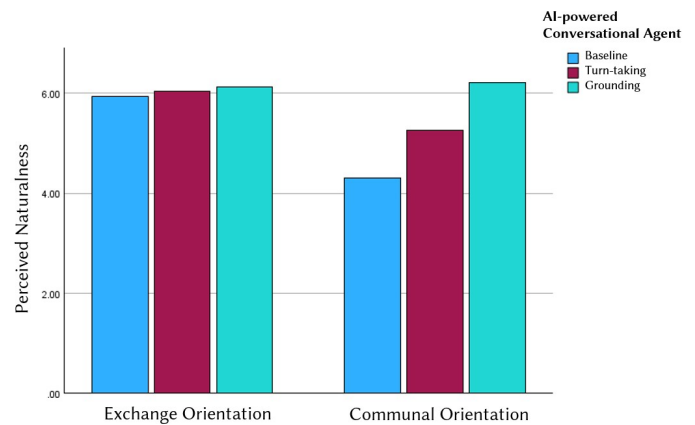


Figure 3: Moderation role of relational orientation (Study 2)

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